

Decentralized versus Learned Swarm Coordination for Communication-Constrained Lunar Surface Exploration

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Abstract

Upcoming lunar surface missions will deploy small teams of rovers that must coordinate autonomously, because Earth–Moon communication latency makes real-time teleoperation impossible and inter-rover radio links are themselves range-limited. NASA’s CADRE technology demonstration, which will place three cooperative rovers and a base station at Reiner Gamma, is the concrete instance of this problem. This work asks a question that such missions must answer at design time: *as the usable communication radius between rovers shrinks, how does a coordination policy learned by reinforcement learning compare against established decentralized swarm heuristics, and which degrades more gracefully when rovers are lost mid-mission?*

We build a reproducible simulation of cooperative exploration over procedurally generated lunar terrain with craters, impassable slopes, rim-rock hazards, and permanently shadowed regions in which solar recharging is unavailable. Rovers have finite battery, finite sensing range, and exchange map knowledge only across a range-limited multi-hop mesh network. Four decentralized policies are compared under identical conditions: greedy nearest-frontier exploration, artificial potential fields, ant-colony-style pheromone stigmergy, and a policy trained with Proximal Policy Optimization and executed independently on board each rover. Because every policy is evaluated on bit-identical terrain within each condition, the experiment is a randomized block design and is analyzed with paired tests, Holm–Bonferroni correction, and a factorial ANOVA whose interaction term directly tests whether the ranking of algorithms depends on communication radius.

Across 180 simulated missions spanning 5 communication radii and 3 rover-failure rates with 4 independently generated terrains per cell, *[results summary pending]*

1 Introduction

1.1 Motivation

Robotic exploration of the lunar surface is moving from single large rovers toward small cooperative teams. NASA’s CADRE (Cooperative Autonomous Distributed Robotic Exploration) technology demonstration will deliver three carry-on-bag-sized, solar-powered rovers and a base station to the Reiner Gamma region aboard the Intuitive Machines IM-3 lander under the Commercial Lunar Payload Services initiative; the rovers are designed to select their own driving targets, maintain formation, and perform distributed multistatic ground-penetrating radar sounding without a human in the loop [1, 2, 3].

The reason for that autonomy is physical. The Earth–Moon round-trip light time is on the order of 2.6s before any ground-station scheduling overhead, which rules out the joystick-style

teleoperation used for early planetary rovers. Coordination must therefore happen locally, between the rovers themselves — and that local channel is not free either. Surface-to-surface radio links between small rovers are power-limited and degrade with distance and terrain occlusion, so at any instant a rover may be able to share its map with some teammates and not others.

This produces the design question studied here. If the inter-rover communication radius is generous, every rover effectively shares one global map and coordination is easy. If it is small, each rover must act on a stale, partial picture of what its teammates have already explored, and the swarm risks redundantly re-covering the same ground. How much does that cost, and does a *learned* coordination policy handle it better than the classical decentralized heuristics that are standard in the multi-robot exploration literature?

1.2 Contributions

1. A reproducible, fully seeded simulation of communication-constrained cooperative lunar exploration, in which terrain, rover placement, and every trial outcome are determined entirely by an integer seed (Section 3).
2. A like-for-like implementation of four decentralized coordination policies — three classical, one learned — that share the same action space, sensing model, and simulation loop, so that measured differences are attributable to decision-making alone (Section 4).
3. A randomized block experimental design with matched terrain across algorithms, analyzed with paired tests, family-wise error correction, and a factorial ANOVA whose interaction term is the direct test of the central hypothesis (Sections 5–6).
4. An explicit account of the train/test separation used for the learned policy, of the simplification used to make multi-agent training tractable, and of two implementation defects found and corrected during development that materially changed the results (Sections 4.5, 9).

1.3 Hypotheses

- H1** The four policies differ in mean final coverage of traversable terrain.
- H2** The performance gap between policies depends on the communication radius R_c ; specifically, differences grow as R_c shrinks. In the factorial model this is an *interaction* between algorithm and R_c , not a main effect.
- H3** The policies differ in how gracefully they degrade when a fraction of the swarm is disabled mid-mission.

The corresponding null hypotheses are that no such differences exist.

2 Related Work

Frontier-based exploration. Yamauchi introduced the frontier concept — the boundary between mapped free space and unknown space — and showed that repeatedly driving to the nearest frontier is sufficient to explore an unknown environment [4]. Burgard et al. extended this to teams by explicitly coordinating which robot claims which frontier, trading off expected information gain against travel cost [5]. The greedy, uncoordinated variant used here as a baseline is deliberately the uncoordinated version, since the point of comparison is how much coordination the other methods add.

Potential fields. Khatib’s artificial potential field formulation treats goals as attractors and obstacles as repulsors, with motion following the resulting gradient [6]. It extends naturally to multi-robot dispersion by treating teammates as repulsors, which is how it is used here.

Stigmergy and ant colony methods. Grassé’s notion of stigmergy — coordination mediated by modification of the shared environment rather than by direct signaling — underpins ant colony optimization [7]. This is a particularly relevant baseline for the present problem because it requires *no radio communication at all*: the coordination channel is the terrain itself.

Learned multi-robot exploration. Recent work applies deep reinforcement learning to multi-robot exploration under restricted sensing and communication, including constrained field-of-view exploration at scale [13], explicit-communication policies with density-based frontier selection [14], frontier-led swarming [15], and heterogeneous teams for planetary analogues [16]; surveys cover the broader area [17]. The present work does not claim to advance that state of the art. Its contribution is a controlled, fully reproducible comparison, at a scale that can be independently re-run on a laptop, of a learned policy against classical baselines under a communication constraint that is varied systematically.

3 Simulation Environment

All quantities below are implemented in `src/lunar_swarm/` and are stated here exactly as coded. The terrain is a stylized model chosen to reproduce the *qualitative* features that constrain rover navigation (bowl-shaped craters, raised rocky rims, impassable slopes, permanently shadowed crater floors); it is not a selenographic reconstruction of a specific site, and no claim is made that its parameters are calibrated to measured lunar data.

3.1 Terrain generation

Terrain is an $S \times S$ grid of unit cells, with one cell treated as 1 m. Everything is a deterministic function of an integer seed.

Base relief. A multi-octave band-limited noise field is formed from $O = 3$ randomly oriented sinusoidal planes,

$$g(x, y) = \sum_{o=0}^{O-1} 2^{-o} \sin\left(\frac{2\pi 2^o}{S} u_o(x, y) + \phi_o\right), \quad u_o(x, y) = x \cos \alpha_o + y \sin \alpha_o, \quad (1)$$

with orientations $\alpha_o \sim \mathcal{U}(0, 2\pi)$ and phases $\phi_o \sim \mathcal{U}(0, 2\pi)$ drawn from the seeded generator. It is rescaled to $\hat{g} = (g - \min g) / \max(g - \min g)$ and the base elevation is $h_0 = 2\hat{g}$ metres.

Craters. For each of K craters, a centre (r_k, c_k) is drawn uniformly from the central 80% of the grid, a radius $\rho_k \sim \mathcal{U}(0.04S, 0.14S)$, and a depth $d_k \sim \mathcal{U}(1.5, 6.0)$ m. Writing $\delta_k(p) = \|p - (r_k, c_k)\|_2$,

each crater contributes a paraboloidal bowl with a raised rim,

$$\Delta h_k(p) = \begin{cases} d_k \left(\left(\frac{\delta_k(p)}{\rho_k} \right)^2 - 1 \right), & \delta_k(p) \leq \rho_k, \\ 0.15 d_k \left(1 - \frac{\delta_k(p) - \rho_k}{0.15 \rho_k} \right), & \rho_k < \delta_k(p) \leq 1.15 \rho_k, \\ 0, & \text{otherwise,} \end{cases} \quad (2)$$

and the final elevation is $h = h_0 + \sum_k \Delta h_k$.

Slope. Slope is obtained from the discrete elevation gradient at unit cell spacing,

$$\theta(p) = \arctan \left(\sqrt{(\partial_y h)^2 + (\partial_x h)^2} \right), \quad (3)$$

expressed in degrees.

Hazards. A cell is a hazard if it is too steep or lies in the rocky rim annulus of any crater:

$$\mathcal{H} = \{p : \theta(p) > \theta_{\max}\} \cup \bigcup_k \{p : \rho_k < \delta_k(p) \leq 1.1 \rho_k\}. \quad (4)$$

A cell is traversable for a rover with slope limit $\theta_{\max}^{\text{rover}}$ iff it is on the grid, $p \notin \mathcal{H}$, and $\theta(p) \leq \theta_{\max}^{\text{rover}}$. (With the default $\theta_{\max}^{\text{rover}} = \theta_{\max} = 25^\circ$ the slope test is implied by $p \notin \mathcal{H}$; it is retained so that rovers with different mobility can be simulated on the same terrain.)

Permanently shadowed regions. Craters deep enough to cast persistent shadow ($d_k \geq 3$ m) contribute a shadowed floor region on the side facing away from a fixed low sun direction $\hat{s} \propto (1, 0.3)$:

$$\mathcal{P} = \bigcup_{k: d_k \geq 3} \left\{ p : \delta_k(p) \leq 0.6 \rho_k \wedge \frac{(p - (r_k, c_k)) \cdot \hat{s}}{\delta_k(p)} < -0.15 \right\}. \quad (5)$$

A rover inside $\mathcal{P}$ receives no solar power, which is the mechanism by which a rover can strand itself. Figure 1 shows an example.

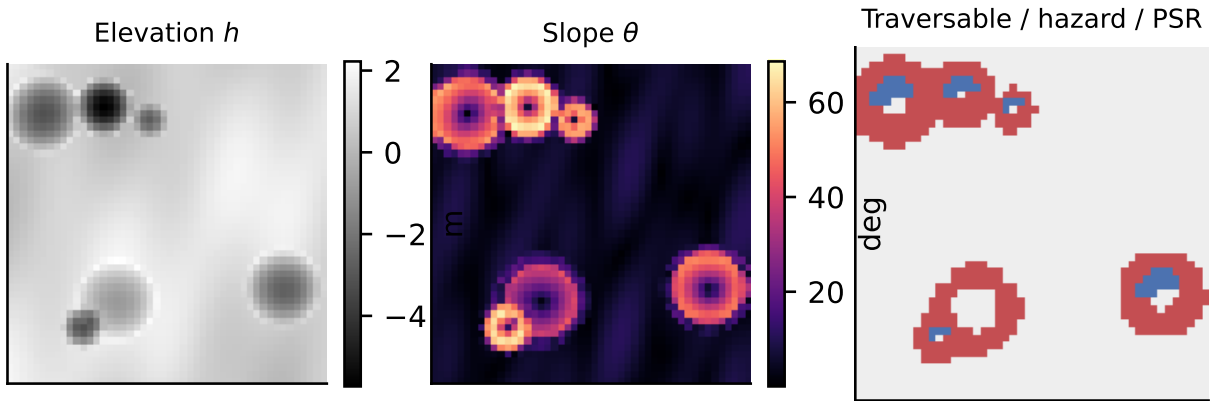


Figure 1: A generated terrain patch (seed 1, $S = 48$, $K = 6$): elevation, derived slope, and the resulting traversability classes. Red marks hazard cells (steep slopes and crater rim rock); blue marks permanently shadowed regions where solar recharge is unavailable.

3.2 Rover model

Each rover i occupies a grid cell p_i and holds a battery level $b_i \in [0, b_{\max}]$ with $b_{\max} = 100$ units. The action set is the eight-connected neighbourhood plus a hold action,

$$\mathcal{A} = \{(\pm 1, 0), (0, \pm 1), (\pm 1, \pm 1), (0, 0)\}, \quad |\mathcal{A}| = 9. \quad (6)$$

Locomotion cost is proportional to ground distance covered,

$$c(a) = \begin{cases} 1, & a \text{ axial}, \\ \sqrt{2}, & a \text{ diagonal}, \end{cases} \quad c_{\text{idle}} = 0.1. \quad (7)$$

A move is executed only if the destination is traversable and $b_i \geq c(a)$; otherwise the rover holds position and is charged c_{idle} , modelling an onboard hazard check that refuses an unsafe command. Rovers in sunlight recharge at $c_{\text{solar}} = 0.6$ units per step, clipped at $b_{\max}$:

$$b_i \leftarrow \min(b_{\max}, b_i - (\text{cost incurred}) + c_{\text{solar}} \mathbb{I}[p_i \notin \mathcal{P}]). \quad (8)$$

A rover with $b_i \leq 0$ after the solar update is permanently disabled.

Crucially, the *energy expended* by rover i is accumulated separately from its battery level,

$$e_i = \sum_t \left(c(a_{i,t}) \mathbb{I}[\text{move executed}] + c_{\text{idle}} \mathbb{I}[\text{held or blocked}] \right), \quad (9)$$

because solar recharge means the final battery level would badly understate how much energy a rover actually used.

3.3 Sensing and knowledge

A rover senses all grid cells within a Euclidean disc of radius R_s and adds them to its known set $\mathcal{K}_i$:

$$\mathcal{K}_i \leftarrow \mathcal{K}_i \cup \{p : \|p - p_i\|_2 \leq R_s\}. \quad (10)$$

Sensing is not occluded by relief; this is a simplification (Section 9). The frontier set of rover i is the set of known cells that touch unknown territory in the four-neighbourhood,

$$\mathcal{F}_i = \{p \in \mathcal{K}_i : \exists p' \in \mathcal{N}_4(p), p' \notin \mathcal{K}_i\}. \quad (11)$$

3.4 Range-limited mesh network

At every step an undirected communication graph is formed over the surviving rovers,

$$E_t = \{(i, j) : \|p_i - p_j\|_2 \leq R_c, i \neq j, \text{ both alive}\}, \quad (12)$$

and knowledge is pooled across each connected component $\mathcal{C}$, so relay is multi-hop:

$$\forall i \in \mathcal{C} : \quad \mathcal{K}_i \leftarrow \bigcup_{j \in \mathcal{C}} \mathcal{K}_j. \quad (13)$$

The communication radius R_c is the primary independent variable. Link quality is binary in range and instantaneous within a component, which is a simplification of real radio behaviour (Section 9).

3.5 Mission outcome measures

Let $\mathcal{E}_t$ be the union of all rovers' known cells up to time t (knowledge already gathered by a rover that later dies still counts, since it was already relayed or recorded). Coverage is measured over traversable terrain only, since hazard cells cannot be driven into:

$$C_t = \frac{|\mathcal{E}_t \setminus \mathcal{H}|}{|\overline{\mathcal{H}}|}, \quad \text{CPE} = \frac{C_T}{\sum_i e_i}, \quad (14)$$

where CPE (coverage per unit energy) rewards mapping more ground per unit of battery drawn. An episode ends when all rovers are disabled, when $C_t \geq 0.995$, or at the step cap T .

3.6 Mid-mission failures

To probe resilience, at the first step with $t \geq \lfloor T/2 \rfloor$ a fraction φ of the swarm, $\lfloor \varphi N \rfloor$ rovers, is selected uniformly at random without replacement from the survivors and permanently disabled. This occurs exactly once per episode and is seeded, so all algorithms experience the same failures on the same terrain.

4 Coordination Algorithms

All four policies are decentralized: each rover computes its own action from its own $\mathcal{K}_i$, its own battery, and locally observable teammate positions. None has access to a global map.

4.1 Shared discrete steering rule

Each policy ultimately produces a desired heading $v \in \mathbb{R}^2$, which must be converted into one of the nine discrete actions. All policies use the same rule: rank the eight moves by cosine similarity to v and take the best-ranked move whose destination is currently traversable, falling back to hold if none is:

$$\text{BTA}(p, v) = \arg \max_{a \in \mathcal{A} \setminus \{(0,0)\}, \mathcal{T}(p+\delta_a)} \frac{v \cdot \delta_a}{\|v\| \|\delta_a\|}, \quad (15)$$

where $\mathcal{T}(\cdot)$ is the traversability predicate. The fallback matters: without it, a policy that always commits to its single best-matching heading becomes permanently wedged the moment that heading points at a hazard (Section 9).

4.2 Frontier (greedy nearest-frontier)

The rover drives toward the nearest cell on its own frontier, with no negotiation with teammates:

$$p^* = \arg \min_{f \in \mathcal{F}_i} \|f - p_i\|_2^2, \quad a_i = \text{BTA}(p_i, p^* - p_i), \quad (16)$$

holding position if $\mathcal{F}_i = \emptyset$. Because the only thing that separates two rovers' targets is any difference in their $\mathcal{K}_i$, rovers that are in radio contact — and therefore share a map — tend to choose the *same* frontier. This is a real property of the uncoordinated heuristic, not an implementation artifact.

4.3 Artificial potential field

The rover follows the sum of an attractive pull toward its nearest $M = 25$ frontiers and repulsion from both teammates and nearby hazards:

$$v_i = k_a \sum_{f \in \mathcal{F}_i^{(M)}} \frac{f - p_i}{\|f - p_i\|^{1.5}} - k_r \sum_{\substack{j \neq i, \text{ alive} \\ \|p_j - p_i\| \leq R_r}} \frac{p_j - p_i}{\|p_j - p_i\|^3} - k_h \sum_{\substack{q \in \mathcal{H} \\ \|q - p_i\| \leq R_h}} \frac{q - p_i}{\|q - p_i\|^3}, \quad (17)$$

with $k_a = 1$, $k_r = 3$, $k_h = 2$, $R_r = 6$, $R_h = 3$, and $a_i = \text{BTA}(p_i, v_i)$. The teammate repulsion term is what disperses the swarm. Note $R_h = 3 < R_s = 4$: hazard repulsion only uses terrain the rover could currently sense, so this policy is not given information the others lack.

4.4 Pheromone stigmergy

A scalar pheromone field τ is deposited on the terrain itself and decays geometrically:

$$\tau_{t+1}(p) = \gamma \tau_t(p) + \Delta \mathbb{I}[p = p_{i,t}], \quad \gamma = 0.98, \quad \Delta = 5. \quad (18)$$

Each rover steers toward the least-attractive-to-revisit cell in its local window, scoring candidates by pheromone, whether it already knows the cell, and distance:

$$p^* = \arg \min_{q \notin \mathcal{H}, 0 < \|q - p_i\| \leq R_e} \left[\tau(q) + \lambda \mathbb{I}[q \in \mathcal{K}_i] + \mu \|q - p_i\| \right], \quad (19)$$

with $R_e = 5$, $\lambda = 20$, $\mu = 0.1$, and $a_i = \text{BTA}(p_i, p^* - p_i)$. The pheromone field is shared by all rovers because it is physically on the ground, exactly as in the biological analogue. This policy therefore coordinates *without using the radio at all*, which makes it an important control: any advantage the other methods show at large R_c must come from communication, and pheromone's performance should be comparatively flat in R_c .

4.5 Learned policy (PPO)

Observation. Each rover forms an egocentric observation from its own knowledge only. An 11×11 patch centred on the rover encodes each cell as

$$P(p) = \begin{cases} -1, & p \notin \mathcal{K}_i \quad (\text{unknown}), \\ 0, & p \in \mathcal{K}_i, p \notin \mathcal{H} \quad (\text{known traversable}), \\ +1, & p \in \mathcal{K}_i \cap \mathcal{H} \quad \text{or } p \text{ off-grid}, \end{cases} \quad (20)$$

concatenated with six scalars: normalized battery $b_i/b_{\max}$; the offset to the nearest surviving teammate within R_c , normalized by R_c (zero if none is in range); the offset to the nearest frontier, normalized by S ; and mission progress t/T . The observation is $121 + 6 = 127$ -dimensional. No global state enters it.

Reward. Let $\Delta_t = |\mathcal{K}_{i,t}| - |\mathcal{K}_{i,t-1}|$ be the growth of the rover's knowledge over the step. Note this includes cells relayed by teammates via Eq. (13), so staying in contact is itself rewarded. The per-step reward is

$$r_t = \underbrace{-0.001}_{\text{time}} + \underbrace{50 \frac{\Delta_t}{|\mathcal{H}|}}_{\text{knowledge gained}} - \underbrace{0.02 \mathbb{I}[a_t \neq \text{hold} \wedge p_{i,t} = p_{i,t-1}]}_{\text{blocked move}} - \underbrace{1.0 \mathbb{I}[\text{rover disabled}]}_{\text{terminal}}, \quad (21)$$

with the episode terminating for that rover if it is disabled.

Training procedure and its simplification. Training a fully self-playing multi-agent policy is a research problem in its own right. We use a standard simplification and state it plainly: during training a *single* learner rover is controlled by the policy while its $N - 1$ teammates execute the frontier baseline. At evaluation time, *every* rover runs an independent copy of the same trained network, so execution is genuinely decentralized and parameter-shared. This means the policy is optimized as a best response to frontier-following teammates and is then deployed among copies of itself — a distribution shift that is a genuine limitation of this study, discussed in Section 9.

Training uses PPO [8] as implemented in Stable-Baselines3 [9], with a two-layer MLP policy ([MISSING]), $\gamma =$ [MISSING], learning rate [MISSING], [MISSING] parallel environments, and [MISSING] environment steps per policy.

Train/test terrain separation. Every training episode generates a *fresh* terrain, with seeds drawn uniformly from $[100000, 10^6)$. All evaluation terrains use seeds $0, \dots, 4 - 1$. The two pools are disjoint by construction, so no result reported here was measured on a map the policy trained on. This separation was introduced after we found that the initial implementation reused a single terrain for essentially all training episodes (Section 9).

Reporting across training seeds. Reinforcement learning outcomes vary across training runs, so a single trained network is weak evidence. We train 0 independent policies with different RL seeds and evaluate all of them. Unless stated otherwise, the “RL” condition in tables and figures is the mean over those policies within each matched block; the spread across them is reported separately in Table ??.

5 Experimental Design

Independent variables. Communication radius $R_c \in \{3, 6, 12, 24, 48\}$ cells; mid-mission failure fraction $\varphi \in \{0\%, 25\%, 50\%\}$; algorithm $\in \{\text{frontier, potential field, pheromone, RL}\}$. A secondary sweep varies swarm size N at fixed R_c .

Dependent variables. Final coverage C_T ; coverage per unit energy (Eq. (14)); total energy expended (Eq. (9)); and number of rovers surviving.

Controlled variables. Terrain size $S = 48$, crater count $K = 6$, sensing radius $R_s = 4$, slope limit 25° , step cap $T = 250$, battery capacity 100, and (in the main sweep) swarm size $N = 4$, which is the configuration the RL policy was trained with. Holding N at the trained value avoids evaluating the learned policy outside its training regime in the primary test; the secondary sweep deliberately violates this to probe generalization.

Blocking. For a given (condition, seed) pair, terrain generation and initial rover placement are deterministic functions of the seed alone. Every algorithm therefore runs on bit-identical terrain from bit-identical starting positions. This is a randomized block design with terrain as the block, and it is verified by an automated test (`tests/test_runner.py`).

Sample size. 4 independently seeded terrains per (algorithm, condition) cell, giving 180 total simulated missions in the main sweep and 60 matched blocks for the paired analyses.

6 Statistical Methods

Because algorithms are evaluated on shared terrain, observations are *not* independent across algorithms and independent-sample tests would be inappropriate — they would also be needlessly weak, since between-terrain variance is large (some randomly generated maps are far more open than others). All analyses are therefore paired/blocked.

Randomized-block ANOVA. With k algorithms and n blocks, letting $\bar{y}_{\cdot j}$ be the mean of algorithm j , $\bar{y}_{i\cdot}$ the mean of block i , and $\bar{y}$ the grand mean,

$$SS_{\text{alg}} = n \sum_{j=1}^k (\bar{y}_{\cdot j} - \bar{y})^2, \quad SS_{\text{block}} = k \sum_{i=1}^n (\bar{y}_{i\cdot} - \bar{y})^2, \quad (22)$$

$$SS_{\text{err}} = SS_{\text{tot}} - SS_{\text{alg}} - SS_{\text{block}}, \quad F = \frac{SS_{\text{alg}}/(k-1)}{SS_{\text{err}}/((k-1)(n-1))}, \quad (23)$$

with partial $\eta_p^2 = SS_{\text{alg}}/(SS_{\text{alg}} + SS_{\text{err}})$. This implementation is verified against `statsmodels`' `AnovaRM` in the test suite.

Pairwise comparisons. Each pair of algorithms is compared with a paired t -test on matched blocks and, because coverage is a bounded proportion and need not be normal, with a Wilcoxon signed-rank test [11] as a non-parametric companion. Effect size is the paired standardized mean difference $d_z = \bar{d}/s_d$ [12].

Multiplicity. With four algorithms there are six pairwise comparisons per metric, so p -values are adjusted by the Holm–Bonferroni step-down procedure [10], controlling the family-wise error rate at $\alpha = 0.05$. This implementation is likewise verified against `statsmodels`.

Testing H2. H2 is a claim about an *interaction*, so it is tested with a factorial ANOVA of coverage on algorithm, R_c , and their interaction. A significant interaction means the gap between algorithms changes with communication radius; a significant main effect of R_c alone would only mean that everyone does better or worse with more communication, which is not the claim.

7 Results

[results pending]

8 Discussion

[discussion pending]

9 Threats to Validity and Limitations

Two defects found during development. Both were found by inspection and testing before any result in this paper was generated, and both are now covered by regression tests. They are reported because each one silently produced plausible-looking but wrong numbers.

1. *Policies could become permanently wedged.* The original steering rule committed to the single best-matching discrete heading. If that heading pointed at a hazard, the move was rejected, the policy recomputed the identical target, and the rover re-issued the identical rejected move for the remainder of the episode. On affected terrains all algorithms returned near-zero coverage. Equation (15) (fall through to the next-best traversable heading) fixes this, and `tests/test_baselines.py` asserts that every baseline reaches non-trivial coverage across multiple seeds.
2. *The learned policy trained on a single map.* Environments were constructed with a fixed configuration seed, so after the first episode every subsequent training episode regenerated the *same* terrain; the policy could memorize one layout, and the evaluation seeds overlapped it. Terrain is now resampled every episode from a training seed pool disjoint from the evaluation seeds (Section 4.5).

Simulation fidelity. The environment is a 2D grid abstraction. There is no wheel–regolith interaction, no vehicle dynamics, no localization error or drift, and sensing is an unoccluded disc rather than a ray-cast field of view — so relief does not block line of sight. Terrain is a stylized generator, not a digital elevation model of a real site, and its parameters are not calibrated against measured lunar data. Results should therefore be read as a controlled comparison *within this model*, not as absolute predictions of on-surface performance.

Communication model. Links are binary within R_c and knowledge is pooled instantaneously across a connected component. Real links degrade continuously with range, suffer terrain occlusion, have finite bandwidth, and incur per-hop latency. A more realistic model would likely widen the gap between methods that depend on communication and the pheromone baseline, which does not use it.

RL training simplification. As described in Section 4.5, the learned policy is trained as a best response to frontier-following teammates but deployed among copies of itself. This mismatch is a genuine confound: it may understate what a properly self-played policy could achieve. Iterative self-play, where teammates are periodically replaced by snapshots of the improving policy, is the natural next step.

Tuning asymmetry. The classical baselines use hand-chosen constants (Eqs. (17) onward) that were set to reasonable values and not systematically optimized, while the RL policy’s parameters are optimized by training. Conversely the RL policy received no hyperparameter search. Neither family has been tuned to exhaustion, and a dedicated tuning effort on either side could move the ranking.

Scope of the swarm-size result. The RL policy was trained at $N = 4$. The swarm-size sweep evaluates it at other N without retraining, so any degradation there reflects out-of-distribution deployment rather than a fundamental limit of the approach.

10 Conclusion

[conclusion pending]

Reproducibility

Every number and figure in this paper is generated from the committed results files by `scripts/make_paper_assets.py`, none is transcribed by hand. The full pipeline is:

```
pip install -r requirements.txt && pip install -e .  
python -m lunar_swarm.rl.train --timesteps 500000 --seed 0 --out-name ppo_seed0  
python scripts/run_paper_experiments.py  
python scripts/make_paper_assets.py  
tectonic paper/paper.tex
```

Environment: Python 3.13.9, NumPy 2.5.3, SciPy 1.18.1, statsmodels 0.15.0, PyTorch 2.14.0, Stable-Baselines3 2.9.0, on arm. Source, trained policies, raw per-trial CSVs, and this document are available at <https://github.com/ViraatC22/lunar-swarm-nav>.

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